

Antonio Carmagnani

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PROFILE

Experience in computational design and data analysis for real estate and cities, spanning academic research and architecture practice. Recently completed an M.Arch. in Architecture Computation at The Royal Danish Academy, with a focus on computational and data methods for real estate and urban development. Hands-on skills span Python, QGIS, and machine learning, parametric workflows in Grasshopper, and building data tools that turn spatial and financial data into decisions. Motivated by work where the built environment, technology, and data intersect.

EXPERIENCE **Project Assistant in International Infrastructure Projects @ Sweco Danmark A/S** **Copenhagen, DK**

02.2026 - Present

- Support consulting teams delivering internationally funded infrastructure projects in water, energy, and transport across emerging and developing economies, financed by development banks including the World Bank, EBRD, IFDK, KfW and EIB.
- Developed a pricing tool to support bid decisions, parametrizing expert fee bands and project economics for financial viability testing of tender configurations.
- Built geospatial databases for water and infrastructure projects in QGIS, compiling georeferenced asset locations from agency data, satellite imagery, and open sources for socio-spatial analysis.
- Engaged national water agencies and government counterparts during the field missions to align on rehabilitation priorities and collect technical and geographic project data.

Intern in Responsible Business Conduct @ Ramboll Management Consulting **Copenhagen, DK**

05.2025 - 08.2025

- Conducted human rights due diligence for corporate clients, applying the UN Guiding Principles on Business and Human Rights and OECD Guidelines to assess salient risks across global supply chains.
- Built a fully automated risk-assessment system in Excel, consolidating risk data across supply chains into a single dashboard that updated salient risks dynamically.
- Integrated AI research tools into due diligence workflows to source and cite human rights risk data at scale across high research volumes.

Architect @ Isay Weinfeld Architects **São Paulo, BR**

02.2023 - 08.2024

- Delivered high-end residential, hotel, and mixed-use projects across South America, North America, Europe, the Middle East, and Asia at one of Brazil's most prominent architecture studios.
- Built a Grasshopper plugin in Python that generates bespoke staircases from a floor plan, cutting staircase modeling from days to around 20 minutes and enabling far more design iterations and geometric complexity.
- Presented the tool in a studio workshop and supported its adoption by colleagues across 10+ projects.

Trainee in Real Estate Product Development @ OSPA **São Paulo, BR**

02.2022 - 10.2022

- Produced 15+ feasibility studies for mixed-use and residential developments across Brazil and the US (São Paulo, Rio de Janeiro, Porto Alegre, Miami) for developer clients.
- Ran early-stage financial feasibility on candidate sites, sizing investment and return against buildable area, unit mix, price per square metre, and parking to test viability.
- Produced images, plans, and diagrams and presented feasibility outcomes to developer clients.

Co-Founder @ VIZU **São Paulo, BR**

02.2022 - 10.2022

- Co-founded and ran an architectural visualization studio, managing 10+ clients, project scoping and delivery, and hiring staff.

Architect @ Felipe Hess Architects **São Paulo, BR**

07.2021 - 02.2022

- Developed high-end residential projects in São Paulo and Rio de Janeiro from preliminary studies through technical documentation, working in Revit, AutoCAD, and SketchUp.
- Took part in client meetings and produced preliminary design studies to align proposals with client briefs.

Intern in Project Management @ Itaú-Unibanco **São Paulo, BR**

08.2020 - 07.2021

- Built a VBA-powered Excel dashboard tracking the bank's corporate furniture inventory across locations, replacing a manual process and speeding up procurement for new-building fit-outs.
- Reviewed technical documentation from architecture and engineering firms prior to construction authorization, checking drawings against building, accessibility, and fire-safety norms (NBR standards).

EDUCATION **M.Arch. in Architecture Computation @ The Royal Danish Academy** **Copenhagen, DK**

07.2024 - 07.2026

Semester Programme in Building Technology @ Architecture Association **London, UK**

07.2022 - 01.2023

06.2019 - 07.2019

07.2018 - 12.2023

PROJECTS

Real Estate Development Recommendation Tool @ The Royal Danish Academy

Copenhagen, DK

- Built a machine learning tool that generates site-specific development recommendations for buildings in Copenhagen by integrating financial, social, sustainability, and geospatial data.
- Designed a user interface allowing different stakeholders to adjust impact drivers and weighting, with predictive outputs aligned to ESG priorities and local development conditions.

Master's Thesis, Computational Urban Analysis for Intensification Strategy @ The Royal Danish Academy

Copenhagen, DK

- Developed a computational framework to identify underused spaces in the historic city center of São Paulo, Brazil, combining urban morphology analysis, land use data, and social vulnerability indicators.
- Built an algorithm that uses urban data, including building typology, use patterns, regulatory conditions, and building use, to generate programmatic and morphological typologies for urban intensification, aligned with the city's existing urban policies.
- Integrated multiple geospatial datasets in QGIS to map spatial relationships between buildings, zoning conditions, and social context at the neighborhood scale.

House Price Prediction - Competition

Copenhagen, DK

- Built a predictive model for real estate valuation using ensemble learning (XGBoost, LightGBM, Ridge, CatBoost) and feature engineering on structured property and financial data, finishing in the top 2% of competitors.

SKILLS

- Data & Analysis: Python, QGIS, SQL, Power BI, Excel / VBA, HTML
- Digital Tools: Rhino + Grasshopper, Revit, AutoCAD, Adobe Suite (InDesign, Illustrator, Photoshop)
- ESG & Sustainability: CSRD, Double Materiality, EU Taxonomy, UNGPs, OECD Guidelines, HRDD, HRIA, LCA (Life Cycle Assessment)
- Real Estate & Urban: Real estate feasibility modeling, Zoning and land use regulatory analysis, Urban morphology and spatial analysis
- Languages: English (fluent), Portuguese (native), Spanish (fluent), Danish (limited working proficiency), Italian (limited working proficiency)